

NISHVI PATEL

Full-Stack AI Engineer | ML Architect | Cloud Infrastructure Specialist

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PROFESSIONAL SUMMARY

Full-Stack AI/ML Engineer & Cloud Infrastructure Architect with 3+ years building end-to-end AI systems, scalable data pipelines, and production cloud infrastructure. Expert in LLM applications, predictive modeling, agentic AI systems, data engineering, Terraform IaC, and AWS cloud architecture. Proven track record designing intelligent automation platforms, deploying ML/AI solutions at scale, architecting microservices infrastructure, and reducing operational overhead by 30-40%. Bilingual technical leader fluent in translating business requirements into data-driven AI solutions deployed on cloud-native platforms.

CORE COMPETENCIES

AI & LLM Engineering: Prompt Engineering, RAG Systems, LLM Fine-Tuning (LoRA/PEFT), Agentic AI (AutoGen, Semantic Kernel), LLM API Integration, Model Evaluation & Benchmarking

Machine Learning & Data Science: Predictive Modeling, Classification, Regression, Time Series Forecasting, NLP, Statistical Analysis, Model Deployment, MLOps

Data Engineering & Analytics: ETL Pipeline Design, Data Warehousing, Star Schema, SQL (Advanced), Data Quality, Automated Reporting, Business Intelligence

Cloud Infrastructure & DevOps: Terraform (IaC), AWS (VPC, EC2, RDS, S3, Lambda, CloudWatch), Docker, GitHub Actions CI/CD, Multi-Environment Deployments

Security & Compliance: IAM Policies, KMS Encryption, Network Security, Data Protection, Cloud Security Best Practices, GDPR/CCPA Compliance

Tools & Platforms: Python (Advanced), SQL, Terraform, Docker, GitHub Actions, AWS CLI, FastAPI, Power BI, Tableau, Git, Linux

EDUCATION

M.S. in Data Science (AI/ML & Cloud Focus)

Stevens Institute of Technology, Hoboken, NJ | GPA: 3.90 | Jan 2025 – May 2026

B.E. in Computer Engineering

Charutar Vidya Mandal University, Gujarat, India | Jul 2020 – Jul 2024

PROFESSIONAL EXPERIENCE

AI/ML Engineer & Infrastructure Architect, Shana AI, Data & Cloud Consultancy LLC, Remote

Jun 2025 – Present

Designed and deployed production agentic AI systems leveraging LLM orchestration (AutoGen, Semantic Kernel) for enterprise automation; reduced manual workflows by 30% while improving accuracy and consistency

Built end-to-end LLM applications integrating RAG, prompt engineering, and multi-turn conversation handling; fine-tuned domain-specific models (LoRA/PEFT) for financial, legal, and technical domains achieving 20%+ performance improvements

Architected production-grade cloud infrastructure using Terraform spanning VPC design, EC2 management, RDS database deployments, and multi-environment CI/CD pipelines (GitHub Actions); enabled 50+ daily deployments with zero-downtime releases

Deployed automated ML pipelines (model training → evaluation → inference) on cloud infrastructure; implemented model monitoring, performance tracking, and automated retraining workflows

Built comprehensive monitoring (CloudWatch) tracking 200+ metrics; implemented security best practices (IAM, KMS, network isolation) achieving 95%+ compliance with cloud security frameworks

Implemented model evaluation frameworks and cost-benefit analysis across LLM providers; optimized infrastructure costs by 35% through intelligent resource allocation and reserved instance purchasing

Senior Data Analyst & ML Engineer, Acorn Universal Consultancy LLP, Gujarat, India

Aug 2023 – Dec 2024

Architected and deployed production ETL pipelines integrating 3 business platforms across 4 countries; reduced manual reporting by 30% while improving data accuracy and governance

Built advanced Power BI dashboards and self-serve analytics platforms enabling C-suite executives to track 50+ KPIs in real-time; delivered executive reporting reducing analysis time by 40%

Developed predictive ML models (time series forecasting, classification) improving forecast accuracy by 15% and identifying high-ROI business segments for targeted growth

Collaborated with Finance, Operations, and Sales teams to design analytics frameworks and translate complex business requirements into data-driven solutions

Data Science Intern, Elecon Engineering Company, Gujarat, India

May 2023 – Jul 2023

Developed predictive models and ETL pipelines; improved forecast accuracy by 20%; built web scraping infrastructure for competitive intelligence

FEATURED AI + DATA + CLOUD PROJECTS

Conversion Core - Production Infrastructure & ML Deployment Platform

GitHub: <https://github.com/Nishviprp/conversion-core>

Architected production-grade AWS infrastructure (Terraform): VPC with security groups, EC2 auto-scaling, Multi-AZ RDS with failover, S3 data lake, CloudWatch monitoring, KMS encryption

Implemented end-to-end CI/CD pipelines (GitHub Actions): automated testing, containerized deployments (Docker), multi-environment promotion, enabling 50+ daily deployments with zero-downtime releases

Deployed ML pipeline infrastructure: automated model training, evaluation, and inference on cloud; implemented model registry, version control, and automated retraining workflows

Built comprehensive monitoring and observability: CloudWatch metrics/logs, health checks, incident alerts, MTTR reduction by 40%

Optimized infrastructure: 35% cost reduction through resource optimization, reserved instances, intelligent auto-scaling

FinSight - LLM-Powered Financial Platform with Cloud Deployment

GitHub: <https://github.com/Nishviprp/FinSightAi>

Built end-to-end LLM application: Gemini API integration for SEC EDGAR analysis, sentiment analysis, risk extraction, forward guidance in <30 seconds

Engineered FastAPI backend with SQLite caching; deployed on AWS EC2 with RDS database, auto-scaling, and CloudWatch monitoring ensuring 99.9% uptime

Implemented NLP pipelines for financial data extraction; applied sentiment scoring, drift detection; deployed live demo reaching 1000+ users on Hugging Face Spaces

CodeGen-LLM Fine-Tuning & Benchmarking | LLM Engineering

GitHub: <https://github.com/Nishviprp/CodeGen-LLM>

Fine-tuned CodeGen-350M using LoRA/PEFT on 5,000+ Python samples; built evaluation harness for HumanEval benchmark; published benchmarking results demonstrating 95%+ performance with 80% computational reduction

PrivaSeal - AI Privacy Pipeline | LLM + Data Engineering

GitHub: <https://github.com/koensantos/privaseal-project>

Engineered modular PII redaction pipeline: YOLOv8 (faces), Presidio NER (text entities), PaddleOCR (document OCR); built Streamlit dashboard; architected GDPR/CCPA governance framework

AWS-Based LMS & Citibike Analytics | Cloud Data Architecture

GitHub: <https://github.com/Nishviprp/-AWS-Based-LMS-Risk-Dependency-Analysis> | <https://github.com/Nishviprp/citibike-data-warehouse>

Designed star schema data warehouse for 50M+ Citibike records; optimized queries achieving <500ms response times; analyzed AWS infrastructure dependencies and risk; built data lake architecture with S3 partitioning and Lambda ETL

TECHNICAL SKILLS

AI/LLM: Prompt Engineering, RAG, LLM Fine-Tuning (LoRA/PEFT), AutoGen, LangChain, OpenAI/Claude/Gemini APIs, HuggingFace Transformers

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LSTM, NLP, Time Series Forecasting

Data Engineering: SQL (Advanced), Python (Pandas, NumPy), ETL Design, Data Warehousing, Star Schema, Power BI, Tableau

Cloud & DevOps: Terraform, AWS (VPC, EC2, RDS, S3, Lambda, CloudWatch, IAM, KMS), Docker, GitHub Actions, CloudFormation

Development: Python (Advanced), SQL, FastAPI, Streamlit, Git, Bash, Linux, REST APIs